

Chapter 1

Solutions to exploration 1.4

1. Null: The probability of the player wearing the red uniform winning is 0.50
Alternative: The probability of the player wearing the red uniform winning is greater than 0.50
2. Null: $\pi = 0.50$, Alternative: $\pi > 0.50$
3. a. $248/457 = 0.543$, b. Let a coin flip represent a match, heads=red wins, tails=blue wins; flip a coin 457 times keeping track of the number of heads; repeat 1000 times recording the number of heads in each set of 457 flips, c. The p-value = 0.03, which gives us strong evidence against the null and in favor of the alternative that the probability of winning for competitors wearing red is greater than 0.50. mean = 0.499, SD = 0.024
4. Null: $\pi = 0.50$, Alternative: $\pi \neq 0.50$ where π = The probability of the player wearing the red uniform winning
5. a. Both tails are shaded, not just one, b. p-value=0.075 (about twice as large as before), c. $0.543 - 0.50 = 0.043$, $0.50 - 0.043 = 0.457$, thus, 0.457 is the comparable value, d. $0.075/0.543/0.457/\text{null hypothesis}$, e. Switching from a one-sided to two-sided test increases the p-value (by approximately twice as much) and, thus, decreases the evidence against the null hypothesis
6. a. If we use 0.57 as the statistic, the p-value is 0.003, b. Smaller, because the statistic (0.57) moved farther away from the null hypothesized value (0.50) than before (0.543), c. The farther the statistic is from the null hypothesized value, the smaller the p-value will be (strength of evidence increases).
7. a. Center = 0.50 (similar to before), variability (SD = 0.03) is larger, b. p-value = 0.058, larger than the original p-value because there is more variability in the null distribution, c. All things being equal as the sample size increases, the strength of evidence increases (p-value decreases).